Christoffel quietness, reduced

Chebyshev sampling at trivial Weyl, the Fejér deformation, and why $Q_k < 1$ is not the floor plate

Stefan Hamann

August 28, 2026

Abstract

Rounds 381 and 382, on the two programme sides, named the same sampling: the μ-orthogonal polynomial squares against ν. This note asks whether that quietness $Q_k := \langle \pi_k^2 \rangle_\nu / h_k(\mu) \le \bar{q} < 1$ is a theorem of the fold geometry (the 3/8-floor, F1 run-lengths, and the cosine grid), not a measurement. What is proved: Chebyshev–Gauss exactness on the equal-weight cosine grid; the polynomial identity $T_k^2 = (1 + T_{2k})/2$; the first-order pairing $\mathrm{FO}_k[-\nu] = \gamma_k(Q_k - Q_{k-1})$; the one-Rayleigh identity $\langle p_k, p_k \rangle_{\mu-\nu} = h_k(\mu)\,(1 - Q_k)$; and that the Chebyshev sampling of ν stays at the mass ratio α under the *trivial* Weyl bound $|S_m| \le \mathrm{mass}(\nu)$. What is *not* proved: the deduction from that geometry to the μ-OP profile Q_k. On FRAME-A $w = 9$, the Chebyshev predictor stays at $\alpha = 0.061$ while the true Q_k grows to 0.389; the Koksma–Hlawka majorant kD^* is 33 at last-twelve and does not close. The two-period adversary of amplitude 2/3 has $Q_k \equiv 0.683 < 1$ and still crosses $\lambda_{\max}(E_{22}) > 1$. The hydra-legal remainder is the pair (Δ, C): the Fejér/d_{arm} deformation $\Delta_k = Q_k - Q_k^T$ (the live term of C_ε), and the coherence factor $\lambda_{\max} = \max Q \cdot (1 + \mathrm{assist})$ (the L^2 remainder of [3]). The Weyl scale of the original candidate is trivial, not PNT, not subconvex, and not RH-equivalent: the Chebyshev–Weyl route never reaches a zeta-quality demand. No RH claim.

Status. Lemmas 1–3 are proved (finite identities). Lemma 4 is Chebyshev–Gauss on a finite cosine grid. Proposition 1 is a finite machine census, not a cofinal law. Proposition 2 is the scale adjudication. Proposition 3 names the reduction. No RH claim. No L^* claim. No G_ε claim.

1 The object

Write π_k for the μ-orthonormal polynomials, $h_k(\mu) = \int \pi_k^2 \, d\mu = 1$, and

$$Q_k := \langle \pi_k^2 \rangle_\nu = \sum_j v_j \, \pi_k(y_j)^2. \tag{1}$$

The same ratio with monic polynomials is identical (machine check: FRAME-A $w = 9$, maxdev $5 \cdot 10^{-15}$). Round 381 records last-twelve $Q \in (0.35, 0.47)$ on MAIN and $Q > 10^3$ on a scramble; the pairing C_ε is $|\mathrm{FO}_k[-\nu]| \le \frac{1}{32}$ on the last twelve. Round 382 records the Christoffel comparison $h_k(w) \ge (1 - \lambda_{\max}(E_{k+1})) h_k(\mu)$ and names the L^2 tail on the flank blocks as the remainder for $n_0 = N - 1$. The mesh $x = \cos(2\pi j/L)$, $L = 4h - 2$, is a construction identity of [1].

The candidate lemma of this round: a geometric bound $Q_k \le \bar{q} < 1$, with $\bar{q}$ read from the 3/8-floor and the cosine-grid equidistribution of ν-angles, would feed both consequences (a) a one-Rayleigh floor plate and (b) C_ε.

2 What the grid sees

Lemma 1 (T_k^2)**.** *The Chebyshev polynomials satisfy* $T_k^2 = (1 + T_{2k})/2$ *as an identity in* $\mathbb{Q}[x]$*. In angular coordinates* $T_k(\cos\theta) = \cos(k\theta)$*, so* π_k^2 *of the Chebyshev reference is a two-mode trigonometric polynomial (frequencies* 0 *and* $2k$*).*

Lemma 2 (first-order pairing)**.** *On a quasi-definite atomic* μ*, writing* $\gamma_k = h_k/h_{k-1}$ *for the monic Jacobi coefficients,*

$$\mathrm{FO}_k[-\nu] = \gamma_k\big(Q_k - Q_{k-1}\big).$$

Exact over $\mathbb{Q}$ *on a finite node set; FRAME-A* $w = 9$*, first twelve, floating-point residual* $2 \cdot 10^{-17}$.

Lemma 3 (one Rayleigh)**.** *For the* μ*-orthogonal polynomial of degree* k*,*

$$\langle p_k, p_k\rangle_{\mu-\nu} = h_k(\mu)\,(1 - Q_k).$$

This is not $h_k(\mu - \nu)$ *and is* not $\lambda_{\max}(E_{k+1}) < 1$*. It is the quadratic form of one vector. On the three-atom* $(3, -2, 3)$*:* $Q_0 = \frac{1}{2}$ *and* $h_0(1 - Q_0) = 2$*.*

Lemma 4 (Chebyshev–Gauss)**.** *On the equal-weight cosine grid* $x_j = \cos(2\pi j/L)$ *of length* $n-1$ *with* $L = 2n$*, the* μ*-orthonormal sampling of the* full *grid is* $Q_k \equiv 1$ *for every* $k < n/2$ *(maxdev* $4 \cdot 10^{-16}$ *at* $n = 32$*). Consequently the Chebyshev partial sum of a subset is a mass-weighted Fourier coefficient of occupancy against the two-mode density of Lemma 1.*

Lemma 5 (total variation)**.** $\mathrm{TV}(\cos(2k\theta), [0, \pi]) = 4k$*. The Koksma–Hlawka / Erdős–Turán majorant of the Chebyshev sampling error is therefore* $O(kD^*)$ *on the circle.*

3 The geometric candidate, measured

Write $\alpha = \mathrm{mass}(\nu)/\mathrm{mass}(\mu)$ and Q_k^T for the Chebyshev predictor $\sum_\nu v(1 + \cos 2k\theta)/\sum_\mu w(1 + \cos 2k\theta)$. Write D^* for the star discrepancy of the ν-angles in $[0, \pi]$ (mapped to $[0, 1]$), and $S_m = \sum_j v_j e^{im\theta_j}$.

Proposition 1 (census)**.** *On the sealed constructors:*

1. *FRAME-A* $w = 9$*:* $\alpha = 0.0611$*, node-fraction* 0.2834*,* $D^* = 0.1832$*, last-twelve* $Q = 0.3891$*, last-twelve* $|\mathrm{FO}| = 0.00978 < \frac{1}{32}$, $\max|Q - Q^T| = 0.335$*,* $kD^* = 33.4$ *at* $k = 182$*,* $|Q - \alpha|/(kD^*) = 0.0089$.
2. *Chebyshev predictor* Q^T *stays in* $(0.055, 0.079)$ *through the last twelve — at* α*, not at* Q*.*
3. *Core-42:* $Q_{\max} \in [0.164, 0.483] < 1$ *(depth* $N - 1$ *for* $N \le 220$*, else* 80*); eleven small-N windows have last-twelve* $|\mathrm{FO}| \le 0.01182 < \frac{1}{32}$.
4. *MAIN-85 star discrepancy* $D^* \in [0.105, 0.212]$*;* χ_3*-42* $D^* \in [0.125, 0.248]$*. (The* 181*-pack builder of [1] was not rerun: it is too expensive for a discrepancy-only census, disclosed. The two programme surfaces MAIN and* χ_3 *already show* $D^* = \Theta(1)$*.)*

The Koksma candidate $Q_k \le \alpha + O(kD^*)$ *does not close: the majorant exceeds* 1 *already at* $k \sim 10$*. The effective ratio* $|Q - \alpha|/(kD^*)$ *is* $o(1)$*, so the coupling is better than generic Koksma, but the error* $Q - \alpha$ *itself still grows to* 0.30*. That growth is the* μ*-OP deformation off Chebyshev, not a discrepancy of* ν*-angles.*

The node-fraction 0.28 versus mass-ratio 0.06 is the Fejér factor $1 - x = 2\sin^2(\theta/2)$: ν occupies a quarter of the nodes but a thinner share of the $(1-x)$-weighted mass. The 3/8-floor of [4] controls gaps, not OP squares.

4 Weyl scale

Proposition 2 (scale)**.** *Let S_m be the ν-weighted exponential sum on the cosine angles.*

1. Trivial. $|S_m| \le \mathrm{mass}(\nu)$. *For the Chebyshev predictor this yields $Q_k^T \le 2\alpha \sim 0.12 < 1$ uniformly in k. Measured $|S_{2k}|/\mathrm{mass}(\nu) \in (0.04, 0.33)$ at $k \ge 1$ on $w = 9$: near-trivial, no extra cancellation.*

2. PNT / subconvex / RH. *Not demanded. The Chebyshev–Weyl route is already closed by (i) and still does not yield the μ-OP lemma, because $Q_k \ne Q_k^T$.*

3. What would close C_ε. *Lemma 2 needs $|\gamma\Delta Q| \le \frac{1}{32}$, i.e. consecutive differences of Q, not a uniform bound on S_{2k}. On $w = 9$ the last-twelve $|\mathrm{FO}| = 0.0098$ is a smoothness fact of the discrete μ-chain, census-grade.*

4. What would close $n_0 = N - 1$. *Even the true $Q_k \le 0.48$ (core census) is insufficient: $\lambda_{\max}(E_k) = \max Q \cdot (1 + \mathrm{assist})$ with assist 1.5–4 on $w = 9$ (at $k = 183$, $\lambda_{\max} = 0.9998$, $\max Q = 0.393$, assist 1.54). The floor plate is the Gram $\lambda_{\max} < 1$ of [3], not the diagonal $Q_k < 1$.*

*The circularity threat — that a needed Weyl bound might be RH-equivalent — does not arise. The honest position of the lemma on the scale trivial $\leftrightarrow$ RH is **trivial**.*

The better-than-Koksma coupling of Leg B is the two-mode Fourier of Lemma 1 (error $= O(|S_{2k}|)$, not a sum of k frequencies) together with the fact that the *live* error is not that Fourier term at all. No explicit-formula input is consumed; the firewall of the probe forbids zero and prime oracles.

5 Adversaries

Scramble. FRAME-A $w = 9$, seed 1: Q_k crosses 1 at $k = 70$; last-twelve $Q = 4.8 \cdot 10^6$. This is the r381 named break, reproduced. Low-degree discrepancy is comparable to MAIN ($D^* \sim 0.24$ versus 0.18 at depth 40). Scramble does not break the angle-discrepancy first; it breaks the μ-OP sampling at high degree.

Two-period, $c = 2/3$. On $S = 81$, $Q_k \equiv \alpha = 0.683 < 1$ for every k (Chebyshev even/odd: Lemma 4 plus equal occupancy). $\lambda_{\max}(E_{22}) = 1.029 > 1$. Amplitude $c = 1$ has $Q_0 = 1.025 > 1$ and kills at h_0. The Q-chain does not see the binding half-filling adversary of [3]. Consequence (a) of the candidate lemma is false as a sufficient condition.

EXT-heavy seven. Windows $\{69, 96, 97, 99, 107, 117, 129\}$ of [3], where $r_{\max}$ reaches 0.879, have $Q_{\max} \in [0.127, 0.221]$ at depth 200 (all < 0.25). Quietness of Q holds at that depth; the flank-mass ratio is a different object.

6 The reduction

Proposition 3 (reduction)**.** *The candidate "$Q_k \le \bar{q} < 1$ from fold geometry" is strictly larger than the pair (Δ, C):*

1. Chebyshev sampling of ν. *SATZ at trivial Weyl (Lemmas 1–4 and Proposition 2(i)). $Q_k^T \le 2\alpha < 1$.*

2. Deformation $\Delta_k := Q_k - Q_k^T$ *of the* μ*-chain off Chebyshev (Fejér Jacobi-*$(0,1)$ *plus* d_{arm}*). This is the live term of* C_ε *via Lemma 2. Census: last-twelve* $|\mathrm{FO}| \le 0.01182$ *on eleven small-*N *core windows; not a cofinal law.*

3. Coherence $\lambda_{\max}(E_k) = \max_j Q_j \cdot (1 + \mathrm{assist})$*. Assist is the off-diagonal of the* ν*-Gram in the* μ*-OP basis — the r285/r382 remainder. Two-period* $c = 2/3$ *is the named witness that* $Q < 1$ *does not control* $\lambda_{\max}$*.*

The original lemma is therefore reduced, not proved and not vacuously refuted: (i) is a theorem, (ii) and (iii) are strictly smaller named remainders, one on each programme side (C_ε *and* $n_0 = N - 1$*).*

The chain of [1, 2, 3] stands with this split in place of a single geometric Q-bound. T1 / COMPOSE unchanged. No RH claim.

7 Machine checks

Numbered lemmas: `verify_christoffel_quiet.py` (same directory). Standalone: T_k^2, TV, trivial Weyl, Chebyshev–Gauss, one-Rayleigh, two-period constancy. Construction: FRAME-A $w = 9$ pins, FO identity, Koksma failure, Weyl scale, scramble kill, two-period $\lambda_{\max}$, EXT $kz = 69$ at depth 200. Discovery census: `christoffel_quiet_probe.py` (`--smoke` 13/13, full 19/19). Exit line: `CHRISTOFFEL QUIET VERIFIED`.

Claim boundary

Finite identities for Chebyshev sampling and Jacobi pairing on a fixed node set, a named reduction of the geometric quietness candidate to (Δ, C), a scramble break, and a two-period witness that $Q < 1$ is not the floor plate. No statement about zeros of $\zeta(s)$, in either direction. The Weyl-scale adjudication is that *no* zeta-quality bound is consumed or required. No RH claim.

References

[1] S. Hamann, *The* V_3'*-lemma, reduced*, standalone note, August 2026, `rh/problem/v3prime_proof.tex`.

[2] S. Hamann, *The* G_ε*-lemma, reduced*, standalone note, August 2026, `rh/problem/g_eps_lemma.tex`.

[3] S. Hamann, *The entry lemma of the pivot band*, standalone note, August 2026, `rh/problem/pivot_entry_lemma.tex`.

[4] S. Hamann, the 3/8-floor of the mean sieve, campaign probe `mean_sieve_floor_probe.py`.